

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	P. Sai Charan Tej	Reg. No.:	192465048
Section / Batch:		Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues.
- Provide a thorough analysis with validated results and performance evaluation.

Question Type	Focus Area	Questions
Industry Problem Solving Task	Focus on Solving optimization problems using dynamic programming and greedy strategies	Q1

SECTION A – Industry Problem Solving Task

A warehouse uses multiple robots to handle tasks such as picking and sorting items. Analyze how task allocation can be optimized using graph algorithms. Identify constraints such as robot capacity and task deadlines. Design a solution for efficient task distribution. Discuss scalability with increasing robots. Evaluate computational complexity. Interpret results in warehouse automation efficiency.

Code of Conduct

I P. Sai Charan Tej(192465048)__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Industry Problem Solving Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%				
Application of Engineering Concepts	25%				
Solution Design	25%				
Use of Tools	15%				
Analysis	10%				
Professionalism	5%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Algorithm: Autonomous Robot Task Allocation

1. **Start.**
2. Read the number of robots n .
3. Read the number of tasks m .
4. For each task, read:
 - Task ID
 - Workload
 - Deadline
5. Store all task details in a list.
6. Sort the tasks based on their **earliest deadline**.
7. Initialize the workload/capacity of every robot to 0.
8. For each task:
 - Check all available robots.
 - Determine which robots can complete the task within its deadline.
 - Among the eligible robots, select the robot with the **lowest current workload**.
9. Assign the task to the selected robot.
10. Update the robot's workload.
11. If no suitable robot is available, leave the task unassigned.
12. Display the tasks assigned to each robot.
13. Display the total number of successfully assigned tasks.
14. **Stop.**

Complexity

- **Time Complexity:** $O(m \log m + mn)$
- **Space Complexity:** $O(m + n)$
- **Approach:** Greedy task allocation with deadline and capacity constraints.

Output:



SIMATS Online Programming Lab

DAA PROGRAMMING

 POTHURAJU SAI CHARAN TEJ
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QUESTIONS

Q1

Autonomous Robot Task Allocation (Graph + Optimization)

100 marks

1/1 answered

Q1 Autonomous Robot Task Allocation (Graph + Optimization) 100 marks

A warehouse uses multiple robots to handle tasks such as picking and sorting items. Analyze how task allocation can be optimized using graph algorithms. Identify constraints such as robot capacity and task deadlines. Design a solution for efficient task distribution. Discuss scalability with increasing robots. Evaluate computational complexity. Interpret results in warehouse automation efficiency.

Your Code

```
1 n = int(input())
2 m = int(input())
3 tasks = []
4 for i in range(m):
5     task_id, workload, deadline = map(int, input().split())
6     tasks.append((task_id, workload, deadline))
7 tasks.sort(key=lambda x: x[2])
8 capacity = [0] * n
9 assignment = [[] for _ in range(n)]
10 for task_id, workload, deadline in tasks:
11     best_robot = -1
12     best_load = float('inf')
13     for r in range(n):
```

Input

```
3
5
1 4 10
2 3 8
```

Output

```
Task Allocation
Robot 1 : [4, 3] Load = 7
Robot 2 : [2, 5] Load = 6
Robot 3 : [1] Load = 4
Total Assigned Tasks = 5
```